

MENORCA AP, Spain

WMO: 083140

Lat: 39.855N

Lon: 4.216E

Elev: 86

StdP: 100.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	5.3	-1.9	3.2	7.3	-0.1	3.8	7.5	14.2	8.7	13.0	8.8	4.5	0	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	31.8	22.1	30.6	22.1	29.6	22.0	24.9	28.4	24.2	27.8	23.6	27.2	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.9	26.8	23.1	18.0	26.4	22.3	17.2	26.0	76.0	28.5	73.3	27.9	70.8	27.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.8	9.5	1.7	33.9	1.7	1.2	0.5	34.8	-0.5	35.5	-1.4	36.1	-2.7	37.0		
(5)			0.5	25.7	1.3	1.4	-0.4	26.7	-1.2	27.5	-2.0	28.3	-3.0	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	11.0	10.6	12.3	14.6	18.0	22.1	25.0	25.5	22.5	19.5	14.9	12.2		
	DBStd	5.71	2.21	2.45	2.25	2.07	2.27	2.38	1.89	1.90	2.17	2.42	2.84	2.43		
	HDD10.0	52	14	20	6	1	0	0	0	0	0	0	2	8		
	HDD18.3	1089	227	217	186	112	34	2	0	0	1	16	105	190		
	CDD10.0	2751	46	37	79	139	247	364	464	479	374	296	151	77		
	CDD18.3	745	0	0	0	1	22	116	206	221	125	53	3	0		
	CDH23.3	5050	0	0	1	2	74	714	1682	1854	617	106	1	0		
	CDH26.7	1339	0	0	0	0	6	163	517	564	86	3	0	0		
(14) Wind	WSAvg	4.3	4.5	4.8	4.7	4.6	4.1	4.0	3.8	3.9	3.9	4.1	4.6	4.4		
(15) Precipitation	PrecAvg	582	57	53	43	44	33	15	4	21	65	85	92	72		
	PrecMax	974	161	186	149	129	107	78	28	141	248	277	251	191		
	PrecMin	275	0	0	0	2	0	0	0	0	0	2	3	5		
	PrecStd	137	37	41	31	32	28	18	7	29	54	59	57	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	17.9	20.8	23.1	27.4	31.9	32.8	33.8	30.1	27.0	22.8	19.0		
		MCWB	14.1	13.7	14.6	15.5	18.6	20.2	21.9	22.3	22.9	21.3	18.5	15.6		
	2%	DB	16.4	16.5	18.7	21.2	25.2	30.1	31.6	32.0	28.7	25.6	21.2	17.9		
		MCWB	13.4	12.9	13.9	15.4	17.7	20.3	21.9	22.7	22.2	20.8	17.8	14.8		
	5%	DB	15.5	15.5	17.4	20.0	23.8	28.6	30.6	30.8	27.6	24.5	20.2	17.0		
		MCWB	12.8	12.4	13.3	15.0	17.5	20.2	21.9	22.9	21.7	20.3	17.1	14.4		
	10%	DB	14.8	14.6	16.2	18.8	22.4	27.1	29.5	29.6	26.5	23.4	19.2	16.1		
		MCWB	12.3	11.8	12.7	14.4	16.9	19.9	21.8	22.6	21.3	19.6	16.4	13.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	14.6	15.8	17.3	20.4	23.7	25.5	26.2	24.9	23.0	19.7	16.2		
		MCDB	16.5	16.7	19.1	20.5	24.1	27.5	29.4	29.8	27.4	25.0	21.4	17.9		
	2%	WB	14.1	13.8	14.7	16.3	19.3	22.5	24.5	25.2	23.9	21.9	18.7	15.5		
		MCDB	15.8	15.6	17.4	19.7	22.8	26.4	28.4	28.7	26.8	24.3	20.5	17.1		
	5%	WB	13.4	13.0	14.0	15.6	18.5	21.6	23.7	24.6	23.1	21.1	17.7	14.9		
		MCDB	15.0	14.7	16.2	18.7	22.0	25.6	27.6	28.1	26.1	23.3	19.7	16.5		
	10%	WB	12.7	12.4	13.3	15.0	17.7	20.9	23.1	24.0	22.3	20.4	16.7	14.1		
		MCDB	14.4	14.1	15.4	17.7	21.2	25.0	27.0	27.5	25.4	22.7	18.9	15.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.4	7.0	7.3	7.7	8.2	8.1	7.7	7.2	6.5	6.1	6.0		
		MCDBR	6.7	7.6	8.5	9.2	9.5	10.2	9.5	9.0	7.9	7.1	6.9	6.8		
	5% WB	MCWBR	4.9	5.3	5.2	5.0	4.3	4.1	3.8	3.7	3.9	4.0	4.7	4.9		
		MCWBR	6.1	6.9	7.4	8.0	8.1	8.6	8.1	7.5	6.9	6.3	6.4	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.348	0.385	0.407	0.422	0.423	0.438	0.428	0.417	0.394	0.363	0.336		
	taud		2.525	2.469	2.364	2.317	2.314	2.348	2.322	2.357	2.376	2.422	2.445	2.503		
	Ebn at Noon		839	869	868	869	860	856	839	839	825	804	784	798		
	Edh at Noon		75	91	112	125	128	124	126	119	110	95	81	72		
(44) All-Sky Solar Radiation	RadAvg		2.08	2.89	4.31	5.58	6.57	7.37	7.26	6.38	4.73	3.37	2.22	1.85		
	RadStd		0.15	0.33	0.34	0.33	0.41	0.31	0.27	0.27	0.21	0.27	0.19	0.13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page